

SEP and Spike-Gate Detection Note

Why the +48 bar-profile spike can be missed by normal SEP while Spike Gate catches it

Summary

The short version is that normal SEP is not looking at the bar-profile peak directly. With the screenshot settings, normal SEP detects objects on the 2D residual image, then applies the configured segment filters.

The Spike Gate path is different: it first finds candidate spike positions in the 1D bar-major profile, then asks a low-threshold SEP pass which 2D segments intersect those spike radii.

Why the +48 Feature Can Be Missed

- The normal SEP pass uses `detect_on=residual`, so SEP sees the residual image rather than the original 1D bar-major profile.
- The normal SEP Detection Threshold in the screenshot is high: `thresh=7.6`.
- A feature can look large in the 1D profile but still fail to become a strong enough 2D residual segment after background/model subtraction.
- SEP may detect the object initially but then reject it through the configured filters: maximum area, maximum elongation, or central exclusion.

Why the -25 Feature Is Different

The -25-ish feature appears to be strong enough in the 2D residual detection image to pass the normal SEP detection and filtering route. That is why it appears in the normal SEP processed profile.

What the Bottom Panel Shows

The SEP & Spike Gate panel does catch the +48 region because that path uses the low-threshold SEP setting, `thresh=0.5` in the screenshot, then keeps only low-threshold SEP segments that overlap radii flagged by the 1D spike detector.

That behaviour is the intended value of the Spike Gate: it can catch profile-damaging spikes that the stricter normal SEP pass misses.

Useful Future Diagnostic

A useful next step would be a small diagnostic readout for each spike candidate with: radius, profile value, neighbour level, side level, SEP label found, and kept/rejected reason. That would make it clear whether a feature was missed by normal SEP thresholding or rejected by a later filter.